```
import pandas as pd
import numpy as np
data = {
    "Customer_ID": ["C101","C102","C103","C104","C105","C106","C107","C108","C109","C110"],
    "City": ["Bengaluru","Mysuru","Hubballi","Bengaluru","Mangaluru",
             "Mysuru","Hubballi","Bengaluru","Mangaluru","Mysuru"],
    "Age": [25, np.nan, 31, 28, 42, 35, np.nan, 29, 38, 27],
    "Income": [45000, 38000, np.nan, 52000, 47000, 41000, 36000, np.nan, 55000, 40000],
    "Membership": ["Gold","Silver","Gold",np.nan,"Silver",
                   "Gold","Bronze","Gold","Silver","Bronze"],
    "Purchase": [3200,2100,4100,3500,2800,3900,1800,4600,5200,2400]
}

df = pd.DataFrame(data)
print(df)
```

```
  Customer_ID       City   Age   Income Membership  Purchase
0        C101  Bengaluru  25.0  45000.0       Gold      3200
1        C102     Mysuru   NaN  38000.0     Silver      2100
2        C103   Hubballi  31.0      NaN       Gold      4100
3        C104  Bengaluru  28.0  52000.0        NaN      3500
4        C105  Mangaluru  42.0  47000.0     Silver      2800
5        C106     Mysuru  35.0  41000.0       Gold      3900
6        C107   Hubballi   NaN  36000.0     Bronze      1800
7        C108  Bengaluru  29.0      NaN       Gold      4600
8        C109  Mangaluru  38.0  55000.0     Silver      5200
9        C110     Mysuru  27.0  40000.0     Bronze      2400
```

```
print("shape:",df.shape)
print("Data types:")
print(df.dtypes)
```

```
shape: (10, 6)
Data types:
Customer_ID     object
City            object
Age            float64
Income         float64
Membership      object
Purchase         int64
dtype: object
```

```
missing_count=df.isnull().sum()
print("Missing values:")
print(missing_count)
```

```
Missing values:
Customer_ID    0
City           0
Age            2
Income         2
Membership     1
Purchase       0
dtype: int64
```

```
missing_percentage=(df.isnull().sum()/len(df))*100
print(missing_percentage)
```

```
Customer_ID     0.0
City            0.0
Age            20.0
Income         20.0
Membership     10.0
Purchase        0.0
dtype: float64
```

```
age_mean=df["Age"].mean()
print("Mean age:",age_mean)
df_mean=df.copy()
print(df_mean)
df_mean["Age"]=df_mean["Age"].fillna(age_mean)

print("Age after mean computation:" )
print(df_mean["Age"])
```

```
Mean age: 31.875
  Customer_ID       City   Age   Income Membership  Purchase
0        C101  Bengaluru  25.0  45000.0       Gold      3200
1        C102     Mysuru   NaN  38000.0     Silver      2100
2        C103   Hubballi  31.0      NaN       Gold      4100
3        C104  Bengaluru  28.0  52000.0        NaN      3500
4        C105  Mangaluru  42.0  47000.0     Silver      2800
5        C106     Mysuru  35.0  41000.0       Gold      3900
6        C107   Hubballi   NaN  36000.0     Bronze      1800
7        C108  Bengaluru  29.0      NaN       Gold      4600
8        C109  Mangaluru  38.0  55000.0     Silver      5200
9        C110     Mysuru  27.0  40000.0     Bronze      2400
Age after mean computation:
0    25.000
1    31.875
2    31.000
3    28.000
4    42.000
5    35.000
6    31.875
7    29.000
8    38.000
9    27.000
Name: Age, dtype: float64
```

```
income_median=df["Income"].median()
print("Median income:",income_median)
df_median=df.copy()
print(df_median)
df_median["Income"]=df_median["Income"].fillna(income_median)
print("Income after median computation:" )
print(df_median["Income"])
```

```
Median income: 43000.0
  Customer_ID       City   Age   Income Membership  Purchase
0        C101  Bengaluru  25.0  45000.0       Gold      3200
1        C102     Mysuru   NaN  38000.0     Silver      2100
2        C103   Hubballi  31.0      NaN       Gold      4100
3        C104  Bengaluru  28.0  52000.0        NaN      3500
4        C105  Mangaluru  42.0  47000.0     Silver      2800
5        C106     Mysuru  35.0  41000.0       Gold      3900
6        C107   Hubballi   NaN  36000.0     Bronze      1800
7        C108  Bengaluru  29.0      NaN       Gold      4600
8        C109  Mangaluru  38.0  55000.0     Silver      5200
9        C110     Mysuru  27.0  40000.0     Bronze      2400
Income after median computation:
0    45000.0
1    38000.0
2    43000.0
3    52000.0
4    47000.0
5    41000.0
6    36000.0
7    43000.0
8    55000.0
9    40000.0
Name: Income, dtype: float64
```

```
import pandas as pd
values=pd.Series([20,25,30,35,100])
print("mean:",values.mean())
print("median:",values.median())
print("mode:",values.mode())
```

```
mean: 42.0
median: 30.0
mode: 0     20
1     25
2     30
3     35
4    100
dtype: int64
```

Double-click (or enter) to edit

```
weather=pd.DataFrame({
    "Day": ["Monday","Tuesday","wednesday","Thursday","Friday"],
    "Temp": [30,np.nan,32,np.nan,31]
```

```
})
print(weather)
```

```
         Day  Temp
0     Monday  30.0
1    Tuesday   NaN
2  wednesday  32.0
3   Thursday   NaN
4     Friday  31.0
```

```
weather_ffill=weather.copy()
print("Before forward fill:")
print(weather_ffill)
weather_ffill["Temp"]=weather_ffill["Temp"].fillna(method="ffill")
print("After forward fill:")
print(weather_ffill)
```

```
Before forward fill:
         Day  Temp
0     Monday  30.0
1    Tuesday   NaN
2  wednesday  32.0
3   Thursday   NaN
4     Friday  31.0
After forward fill:
         Day  Temp
0     Monday  30.0
1    Tuesday  30.0
2  wednesday  32.0
3   Thursday  32.0
4     Friday  31.0
/tmp/ipykernel_1640/3164516030.py:4: FutureWarning: Series.fillna with 'method' is deprecated and will raise in a future version
  weather_ffill["Temp"]=weather_ffill["Temp"].fillna(method="ffill")
```

```
copydf=df.copy()
print(copydf)
copydf["Age"]=copydf["Age"].fillna(copydf["Age"].mean())
print(copydf["Age"])
copydf["Income"]=copydf['Income'].fillna(copydf["Income"].median())
print(copydf["Income"])
```

```
  Customer_ID       City   Age   Income Membership  Purchase
0        C101  Bengaluru  25.0  45000.0       Gold      3200
1        C102     Mysuru   NaN  38000.0     Silver      2100
2        C103   Hubballi  31.0      NaN       Gold      4100
3        C104  Bengaluru  28.0  52000.0        NaN      3500
4        C105  Mangaluru  42.0  47000.0     Silver      2800
5        C106     Mysuru  35.0  41000.0       Gold      3900
6        C107   Hubballi   NaN  36000.0     Bronze      1800
7        C108  Bengaluru  29.0      NaN       Gold      4600
8        C109  Mangaluru  38.0  55000.0     Silver      5200
9        C110     Mysuru  27.0  40000.0     Bronze      2400
0    25.000
1    31.875
2    31.000
3    28.000
4    42.000
5    35.000
6    31.875
7    29.000
8    38.000
9    27.000
Name: Age, dtype: float64
0    45000.0
1    38000.0
2    43000.0
3    52000.0
4    47000.0
5    41000.0
6    36000.0
7    43000.0
8    55000.0
9    40000.0
Name: Income, dtype: float64
```

```
membership_mode=df["Membership"].mode()[0]
print("Mode of Membership:",membership_mode)
copydf["Membership"]=copydf["Membership"].fillna(membership_mode)
print("Missing membership values:")
```

```
print(copydf["Membership"].isnull().sum())
```

```
Mode of Membership: Gold
Missing membership values:
0
```

```
encoded_df=pd.get_dummies(
   copydf,
    columns=['City',"Membership"],
    dtype=int
)
encoded_df
```

	Customer_ID	Age	Income	Purchase	City_Bengaluru	City_Hubballi	City_Mangaluru	City_Mysuru	Membership_Bronze	Membersh
0	C101	25.000	45000.0	3200	1	0	0	0	0	
1	C102	31.875	38000.0	2100	0	0	0	1	0	
2	C103	31.000	43000.0	4100	0	1	0	0	0	
3	C104	28.000	52000.0	3500	1	0	0	0	0	
4	C105	42.000	47000.0	2800	0	0	1	0	0	
5	C106	35.000	41000.0	3900	0	0	0	1	0	
6	C107	31.875	36000.0	1800	0	1	0	0	1	
7	C108	29.000	43000.0	4600	1	0	0	0	0	
8	C109	38.000	55000.0	5200	0	0	1	0	0	
9	C110	27.000	40000.0	2400	0	0	0	1	1	

```
print("Misssig values:")
print(encoded_df.isnull().sum())
print("\n DAta types:")
print(encoded_df.dtypes)
print("Final shape:",encoded_df.shape)
```

```
Misssig values:
Customer_ID          0
Age                  0
Income               0
Purchase             0
City_Bengaluru       0
City_Hubballi        0
City_Mangaluru       0
City_Mysuru          0
Membership_Bronze    0
Membership_Gold      0
Membership_Silver    0
dtype: int64

 DAta types:
Customer_ID           object
Age                  float64
Income               float64
Purchase               int64
City_Bengaluru         int64
City_Hubballi          int64
City_Mangaluru         int64
City_Mysuru            int64
Membership_Bronze      int64
Membership_Gold        int64
Membership_Silver      int64
dtype: object
Final shape: (10, 11)
```

Start coding or generate with AI.